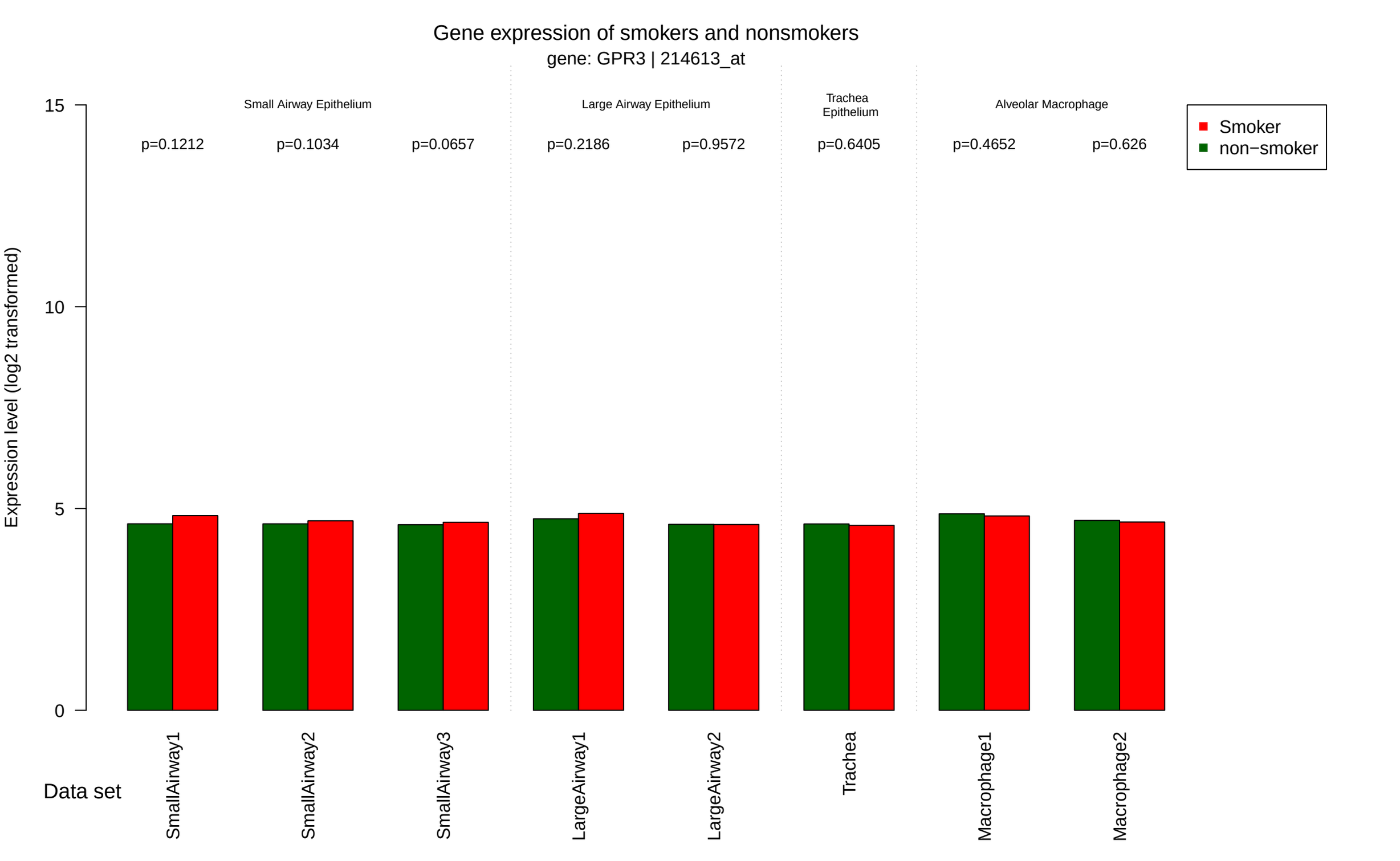
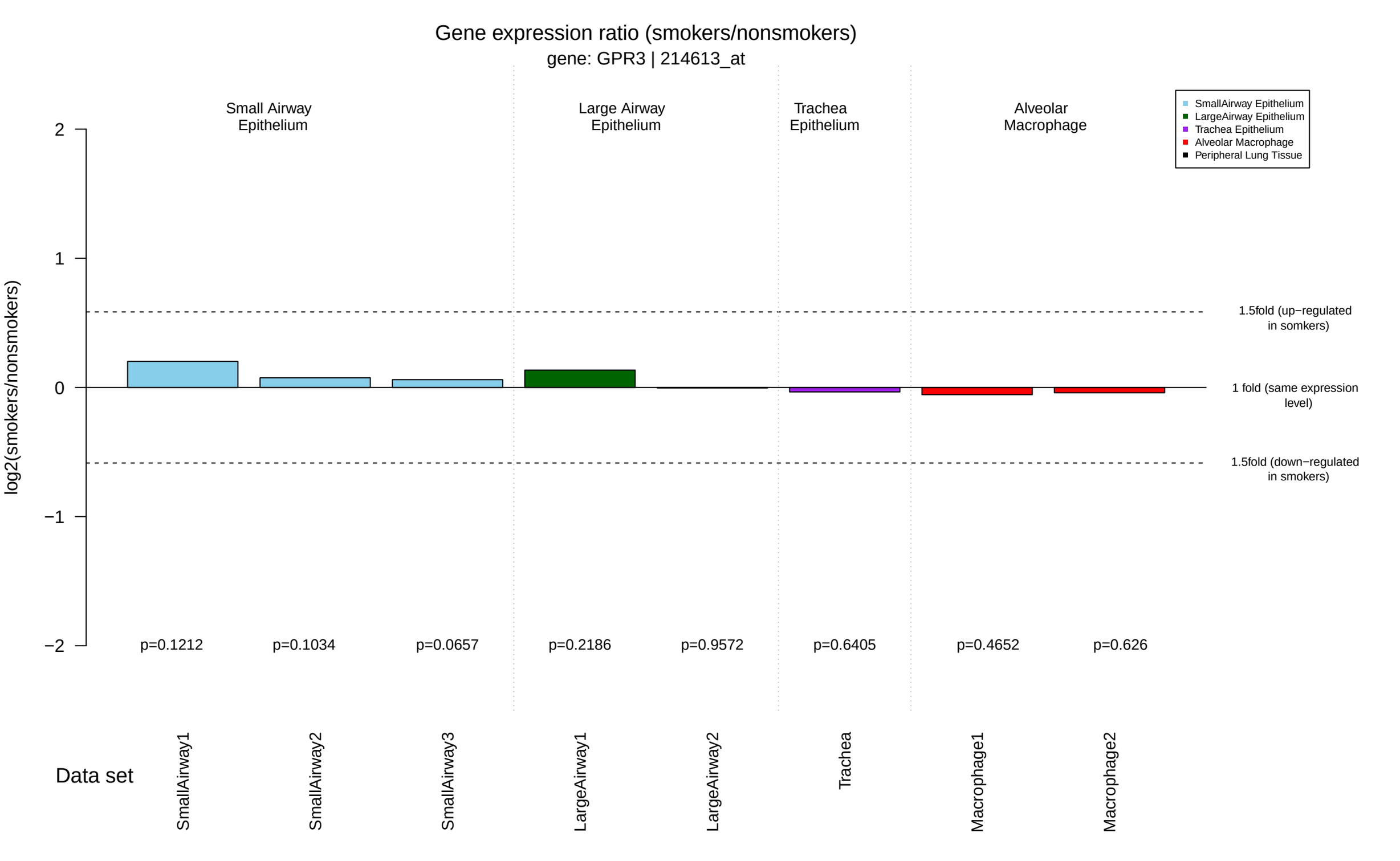
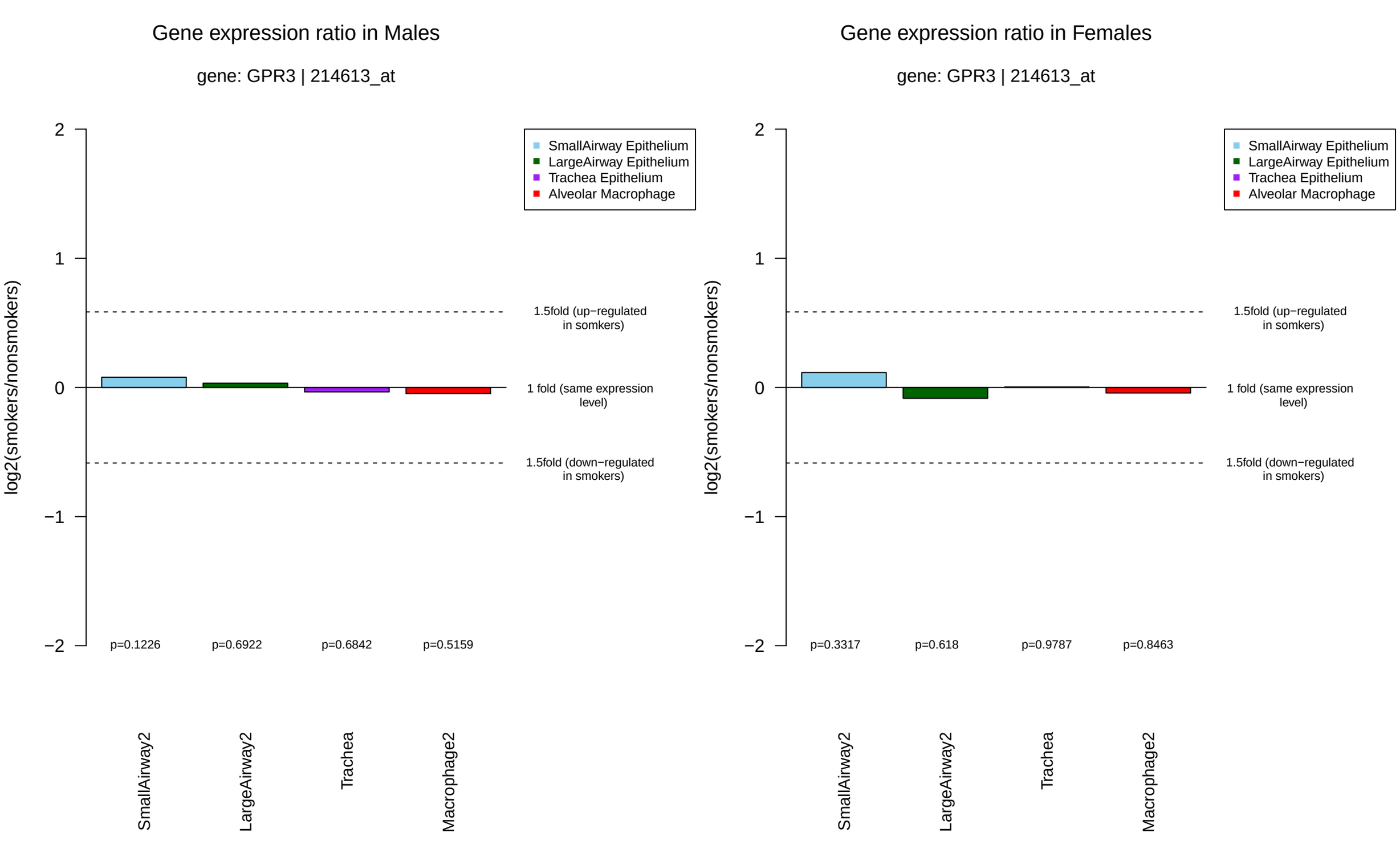
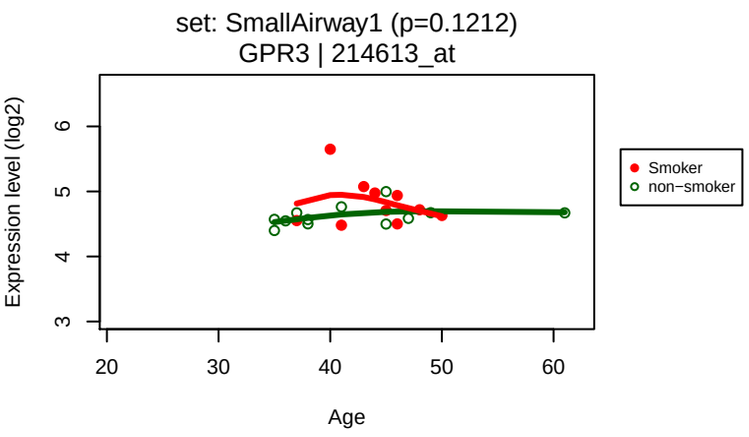
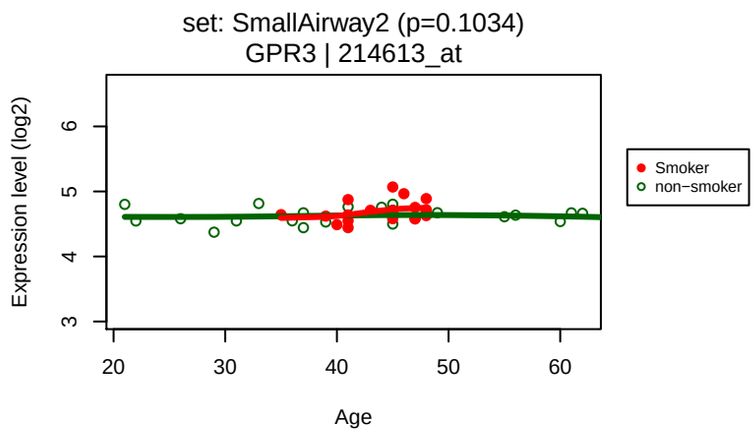
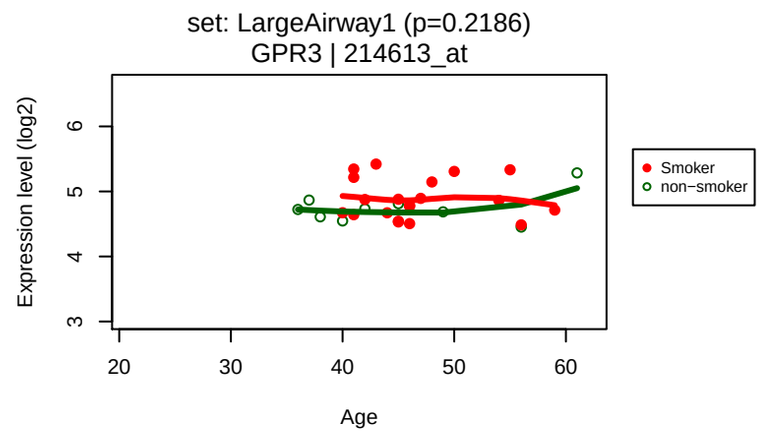

Gene expression of smokers and nonsmokers
gene: GPR3 | 214613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1212
p=0.1034
p=0.0657
p=0.2186
p=0.9572
p=0.6405
p=0.4652
p=0.626
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

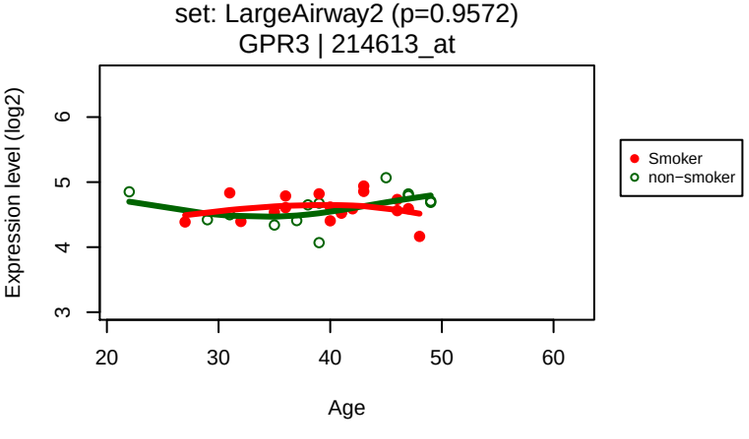

Gene expression ratio (smokers/nonsmokers)
gene: GPR3 | 214613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1212
p=0.1034
p=0.0657
p=0.2186
p=0.9572
p=0.6405
p=0.4652
p=0.626
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

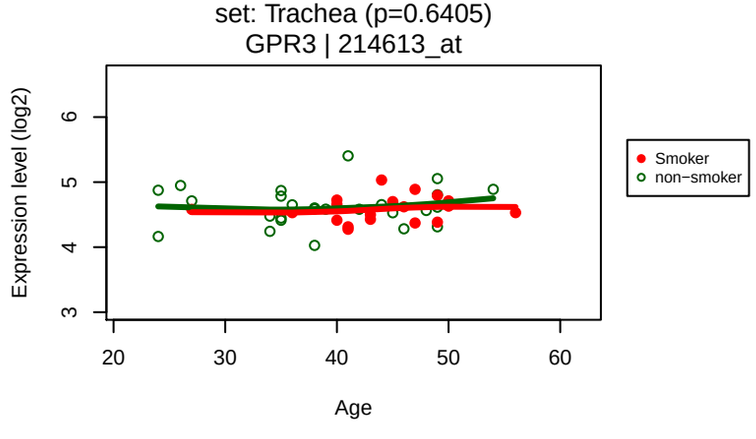

Gene expression ratio in Males
gene: GPR3 | 214613_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1226
p=0.6922
p=0.6842
p=0.5159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR3 | 214613_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3317
p=0.618
p=0.9787
p=0.8463
SmallAirway2
LargeAirway2
Trachea
Macrophage2

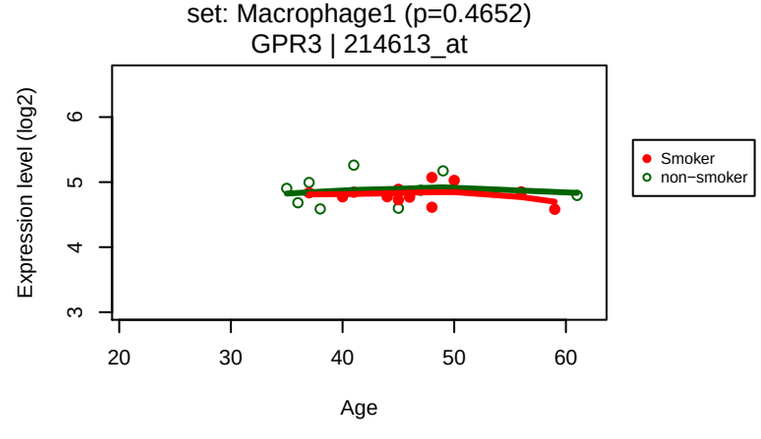

set: SmallAirway1 (p=0.1212)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker

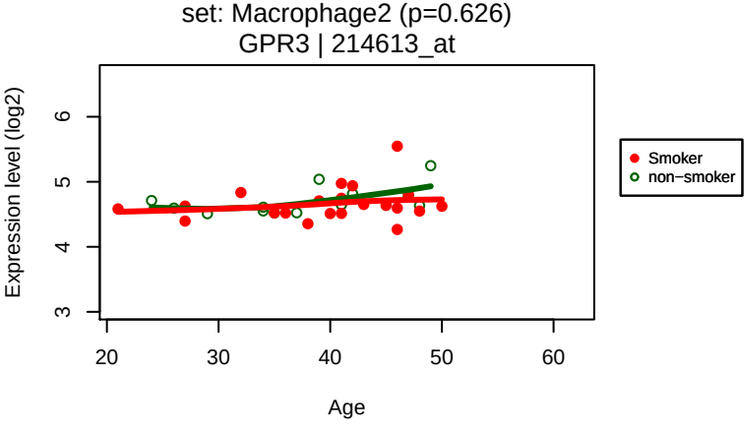

set: SmallAirway2 (p=0.1034)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker

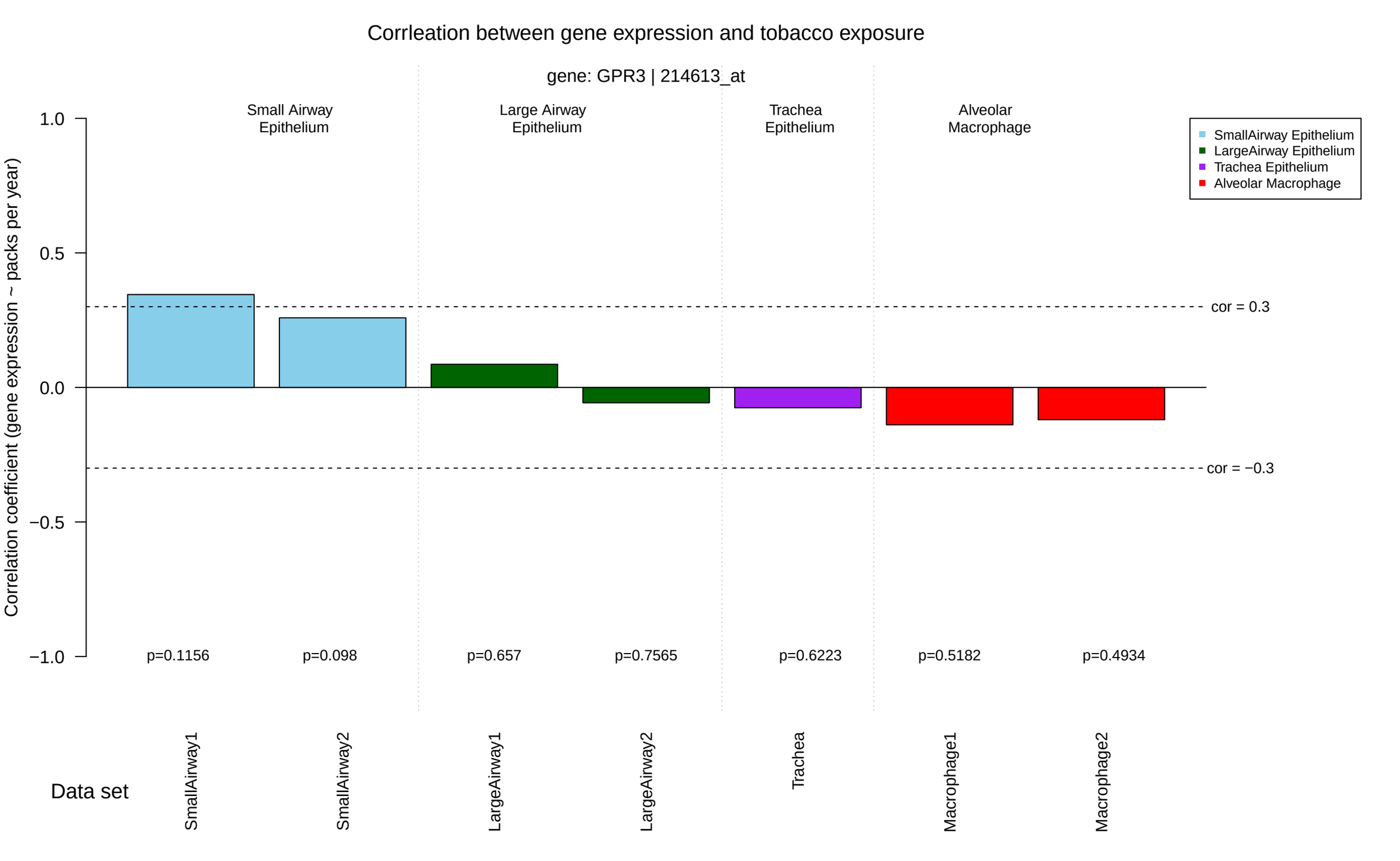

set: LargeAirway1 (p=0.2186)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9572)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6405)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4652)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.626)
GPR3 | 214613_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR3 | 214613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1156
p=0.098
p=0.657
p=0.7565
p=0.6223
p=0.5182
p=0.4934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2